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A Hybrid Attention Mechanism and RepGFPN Method for Detecting Wall Cracks in High-Altitude Cleaning Robots

Haiqiao Liu, Lingding Li, Ya Li, Qing Long, Zhuoyu Chen

Abstract—Aiming at the problem that cracks with different shapes and scales on the exterior walls of high buildings are difficult to detect, this paper proposed a wall crack detection method for high-altitude cleaning robots by hybridizing the GAM attention mechanism and RepGFPN. First, the GAM attention mechanism was incorporated into the YOLOV5 backbone network to reduce information and amplify global features to improve the accuracy of feature extraction. Then, the neck network incorporated the RepFPN method to improve the descriptive ability of fused multi-scale features and to increase computational efficiency. Public datasets Concrete Crack Images for Classification, Mixed VOC2007, CrackForest-dataset-master, and UCMerced_LandUse were used for experimental validation. The ablation experiment results show that the average accuracy of mAP is improved by 13.5 % after introducing the GAM attention mechanism under the yolov5 s original model, while the method in this paper (GR-YOLO) continues to improve by 4.7 %. The experimental results show that the average accuracy mAP of the proposed method (GR-YOLO) is 24.0 %, 47.1 % and 41.0 % higher than that of the model yolov5s + involution, yolov5s + p2 + involution and yolov5s + p2 + involution + CBAM, respectively. The method proposed in this article can more effectively improve the accuracy of crack detection and has important application prospects.

Index Terms—Wall defect; Crack detection; Deep learning; Attention mechanism; Special clinic pyramid

I. INTRODUCTION

With the development of human society and the progress of science and technology, there are more and more high buildings. Wall cracks will not only affect the appearance of buildings and reduce their service life, but more seriously, falling objects on the facade caused by cracks may cause safety accidents, resulting in casualties and irreparable property losses [1]. High-efficiency and rapid crack detection technology is crucial for ensuring the safe operation of buildings and other high-rise structures. The importance is heightened by the prominent aging of global infrastructure and

the continuous emergence of new high-rise structures. Currently, common crack detection methods are mainly divided into two categories, one is based on traditional target detection methods, and the other is based on deep learning detection methods.

The traditional object detection algorithm is a kind of object detection method that realizes binary classification by designing a classifier. In 2004, Paul Viola and Michael Jones proposed the Viola-Jones (VJ) detector [2], which realized real-time face detection in 2017 [3,4]. This traditional algorithm was simple in structure and did not require a lot of data and models but had limited accuracy. In literature [5,6], real-time detection accuracy has been improved to a certain extent by providing the number of samples and the parameter scale of the model. However, the fitting accuracy of the model restricted the accuracy of such methods to a large extent, and the accuracy of such algorithms was greatly reduced in the case of multi-scale and large shape changes.

The deep learning target detection algorithm achieves target feature extraction through convolution, and has a good detection effect on complex scenes and targets with various shape changes. In recent years, with the rapid development of artificial intelligence [7,8], deep learning [9,10], drones [11,12,13] and robots [14], crack detection and recognition technology has become increasingly mature. [15,16,17].

Object detection algorithms based on deep learning include YOLO (You Only Look Once) [18], a single-stage regression algorithm represented by single multiple detectors (SSD) [19], and a two-stage region suggestion algorithm represented by block convolutional neural network (R-CNN) [20]. Kisu Lee et al. [21] proposed a defect detection system for building exterior walls based on Faster-R-CNN. The system could detect many kinds of defects, but its real-time performance was poor. Lu et al. [22] proposed a style transfer method for ceramic tile texture generation that combines classifier-guided StyleGAN and AdaIN-GAN to improve picture quality.

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Since single-stage algorithms usually complete classification and localization in the same stage, they may not be able to locate the bounding box of the target as accurately as the two-stage algorithm. When dealing with small objects, the single-stage algorithm may be difficult to detect accurately due to the lack of feature extraction. To improve the detection accuracy, the model of the single-stage algorithm may become very complex, which increases the computational burden of the model and the risk of overfitting. The two-stage algorithm needs to first perform regional proposals and then classify and locate each proposal. This process is relatively slow and is not suitable for applications with high real-time requirements. Since the two-stage algorithm needs to process more candidate regions, it consumes a lot of computing resources and storage space. In the case of complex background or severe occlusion, the two-stage algorithm may be difficult to effectively separate the target and background, resulting in a decrease in detection performance.

YOLOv5 simplifies the object detection task into an end-to-end single-stage regression problem, utilizing an optimized network architecture (such as CSPDarknet53), multi-scale prediction mechanisms (FPN and PAN), advanced data augmentation techniques (such as Mosaic and MixUp), and regularization techniques (such as label smoothing and CIOU loss). This approach improves detection speed while maintaining high accuracy, resulting in an efficient, accurate, and easily deployable object detection system for real-time applications. Idjaton et al. [23] combined the latest YOLO network and transformer encoder layer based on YOLOv5 to automatically detect the peeling area of limestone walls, with better performance than the YOLO core network. Liu et al. [24] proposed an efficient building damage detection method combining ground images and improved YOLOv5. J Li et al. proposed a new lightweight model based on the YOLOv5 framework, called LEF-YOLO [25]. It was used to detect four extreme wildfires in real-time. Aiming at the problem that the existing small target detection model could not completely extract the features of small targets in the image due to the small coverage area, low resolution, and unclear detail information of small targets. Peng et al. [26] proposed a simple and efficient YOLOv5 algorithm to strengthen the feature pyramid network R-FPN.

In the target detection task, the detection of small objects has always been a challenge. YOLOv5 may have missed detection or inaccurate positioning when dealing with small objects, which is due to the lack of detailed information extraction of its feature extractor on small-scale objects. Moreover, when YOLOv5 faces complex scenes in the real world, its generalization ability may be challenged. For example, in the case of illumination changes, perspective changes, occlusion, etc., the detection performance of the model may be significantly reduced. Although YOLOv5 has great potential in the detection of appearance defects, there is little research on the detection of building facade defects, and the speed and accuracy of the detection of cracks with different shapes and scales need to be improved. Currently, there are few practical

applications of YOLOv8 in crack detection. Moreover, the experimental results in this paper show that the accuracy of YOLOv8 is significantly lower than that of YOLOv5.

To solve the above problems, this paper proposed a method for wall crack detection of high-altitude cleaning robots with a hybrid attention mechanism GAM and RepGFPN. For the convenience of writing, this method is named as GR-YOLO method. Specific innovations are as follows:

(1) To enhance inter-dimensional information interaction, the Global Attention Mechanism (GAM) was integrated into the backbone network of YOLOv5 to effectively capture significant features across all three dimensions and ultimately improve the accuracy of feature extraction.

(2) The combination of RepFPN (Reparameterized Generalized-FPN) in the neck network enabled the GR-YOLO method to achieve real-time, cross-scale feature fusion while ensuring high precision and reducing computational costs.

The structure of the paper is as follows: The first section covers related research; the second section introduces the GR-YOLO method; the third section presents the experimental validation and discussion; the fourth section discusses the limitations of the proposed method and outlines future research directions and the final section concludes the paper.

II. GR-YOLO METHOD

The GR-YOLO method network architecture is shown in Fig. 1. The method was mainly composed of four parts, namely Input, Backbone, Neck, and Head. The first part was the image input part.

The second part was the Backbone, which was mainly responsible for feature extraction of input images and was composed of Conv, C3, GAMAttention, and SPPF modules. The Conv module was a basic module commonly used in convolutional neural networks, mainly composed of the convolutional layer, BN layer, and activation function. The C3 module was composed of three Conv blocks, whose main function was to increase the depth and receptive field of the network and improve the ability of feature extraction. The GAMAttention module was an attention mechanism module that integrated the GAM attention mechanism to enhance the feature extraction ability of wall cracks. The SPPF module was a kind of pooling module, whose function was to realize the spatial invariance and position invariance of input data to improve the recognition ability of neural networks.

The third part was the Neck, which was mainly responsible for the multi-scale feature fusion of the feature map and the transmission of these features to the prediction layer. It consisted of Conv, Concat, Upsample, and CSPStage modules. The Concat module was an operation that concatenated two or more tensors in a certain dimension. The Upsample module was an upsampling operation. The CSPStage module was a feature fusion module combined with RepFPN. The fourth part was the Head, which was mainly responsible for making the final regression prediction. The Head was the part used for object detection of the feature pyramid, which included some convolution layers, pooling layers, and fully connected layers.

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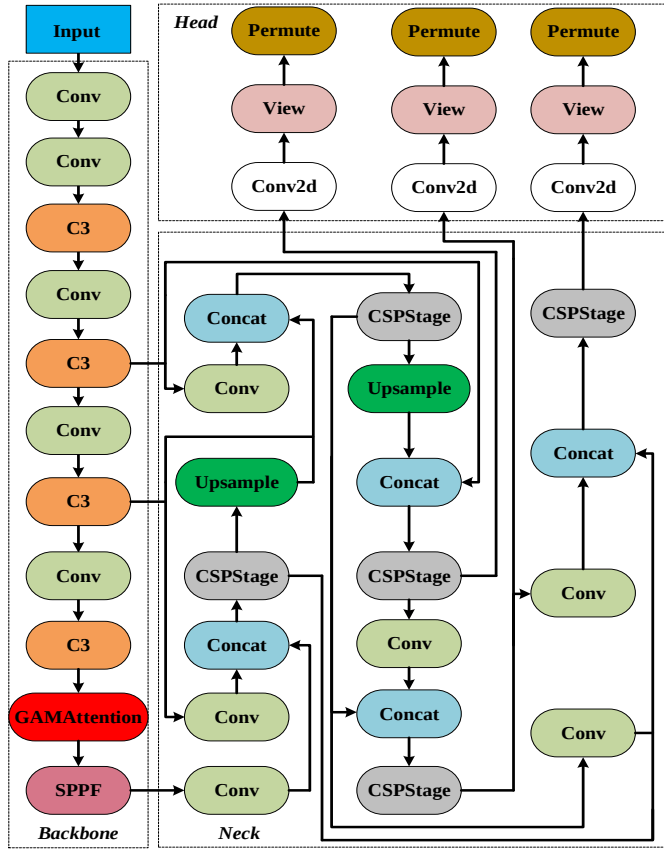


Fig. 1. The overall structure of the GR-YOLO method.

A. Backbone network integrating GAM attention mechanism

To reduce information loss and enhance global dimension interaction features. In this paper, the GAM [27] attention mechanism was adopted in the backbone network of YOLOv5. This mechanism took the sequential channel-space attention mechanism from CBAM [28] and redesigns the submodules. The structure of the GAM attention mechanism is shown in Fig. 2.

In Fig. 2(a), a flowchart of GAM is shown, which contains two main modules, the channel attention module and the spatial attention module. The process of GAM function transfer is described in Equations 1 and 2.

$$f_2 = m_c(f_1) \otimes f_1 \quad (1)$$

$$f_3 = m_s(f_2) \otimes f_2 \quad (2)$$

where f_1 are the input features. f_2 is an intermediate quantity through the channel attention module. f_3 is output features. m_c and m_s are channel and spatial feature maps respectively. $\otimes$ Represents element-by-element multiplication.

The channel attention module, shown in Fig. 2(b), is an important mechanism used to enhance the information interaction between channels in deep neural networks. For the input feature map, dimensionality transformation was first carried out to convert it into a single feature vector, and then the feature vector was processed and modeled by a multi-layer perceptron (MLP) module to obtain the correlation information between channels. Subsequently, the feature vector after MLP

processing was converted back to the original dimension and processed by the Sigmoid function to obtain the attention weight between channels, which was used to adjust the importance of each channel in the subsequent feature representation.

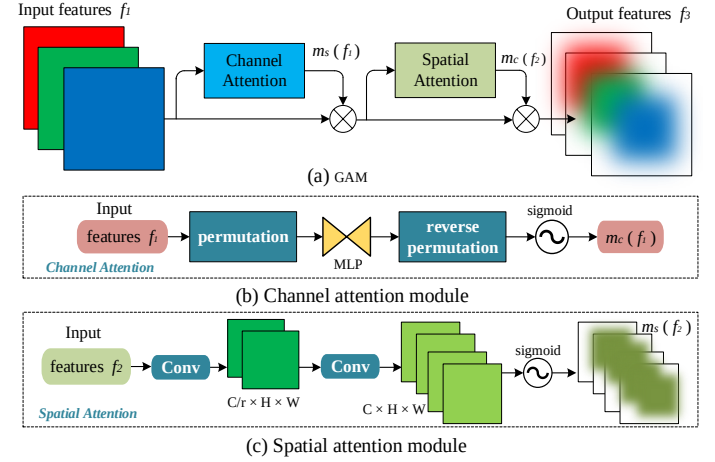


Fig. 2. GAM structure diagram.

As shown in Fig. 2(c), the spatial attention module is mainly used to extract the importance of different spatial positions in the feature map to enhance the network's attention to spatial information. Firstly, by using the convolution operation with convolution kernel 7, the number of channels in the feature graph was compressed to reduce the computation. Then, through another convolution operation with convolution kernel 7, the number of channels was restored to the original state to maintain the consistency of the number of channels. Finally, after processing by the Sigmoid function, the output attention weight was used to adjust the importance of different spatial positions in the feature map, so that the network could use the information of different positions more accurately.

B. RepGFpN-based neck network

To better integrate multi-scale features, and improve the real-time performance of the model, The Efficient-RepGFpN network was used in the neck network [29,30]. The structure of Efficient-RepGFpN is shown in Fig. 3.

In the bottleneck part of feature fusion, as shown in Figure 3, Efficient-RepGFpN uses different scale feature mappings with different channel sizes. This was because the feature maps of different scales had great differences in computational complexity. Under the constraint of limited computational cost, it was difficult to ensure that all scale feature maps shared the same channel dimension. By flexibly controlling the number of channels of different scale feature maps, higher accuracy could be obtained compared to sharing the same channel.

GFpN enhanced the interaction between features through post-fusion and introduced many additional up-sampling and down-sampling operations. Under the constraint of maintaining real-time detection, efficient-RepGFpN eliminated the additional up-sampling operation in bee colony fusion, thereby improving the real-time performance of the model.

In the feature fusion block, efficient-RepGFpN first replaces the original feature fusion based on 3x3 convolution with

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CSPNet (as shown in Fig. 3 (a)) to obtain higher mAP gain. Subsequently, by introducing a re-parameterization mechanism and an efficient Layer Aggregation Network (ELAN) connection, CSPNet was further upgraded to achieve higher accuracy without adding a huge computational burden.

Through these improvements, efficient-RepGFPN could provide more efficient and accurate feature fusion while maintaining real-time detection, thereby obtaining better performance in target detection tasks.

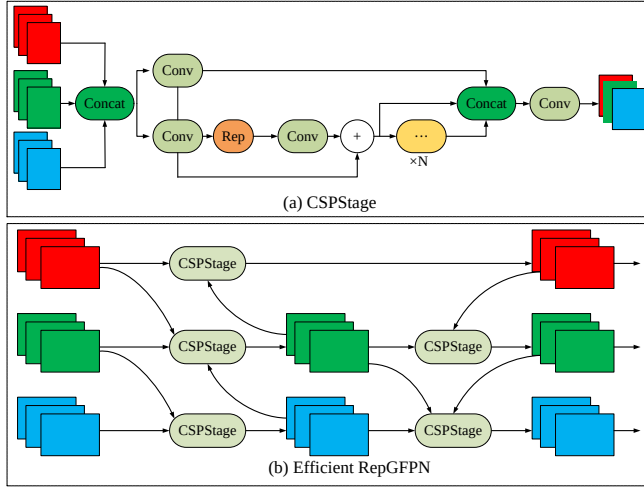


Fig. 3. Efficient-RepGFPN structure diagram.

III. EXPERIMENTAL RESULTS

A. Preparation for Experiment

To enhance the verification of the method proposed in this paper, a verification experiment was carried out. The experimental data are the public data sets Concrete Crack Images for Classification, Concrete-voc2007, CrackForest-dataset master, and used-land use. In this paper, a small sample method was used to select 64 images from the above dataset, including high-altitude wall cracks, wall cracks, and some road cracks, and the images were pre-processed by cutting, rotating, and mapping. Using the annotation tool for manual annotation, 64 images containing crack targets were finally obtained. Part of the training data is shown in Fig. 4.

The experimental environment parameters and hyperparameters are shown in Table I. This includes details such as the operating system version, hardware configuration, input image size, training epochs, batch size, initial learning rate, warmup initial momentum, and optimizer weight decay.

Accuracy P , recall rate R , and average accuracy mAP were used as evaluation indexes. The specific formula is as follows.

$$p = \frac{T_p}{T_p + F_p} \times 100\% \quad (3)$$

$$R = \frac{T_p}{T_p + F_n} \times 100\% \quad (4)$$

$$mAP = \frac{\sum_{i=1}^n AP_i}{n} \times 100\% \quad (5)$$

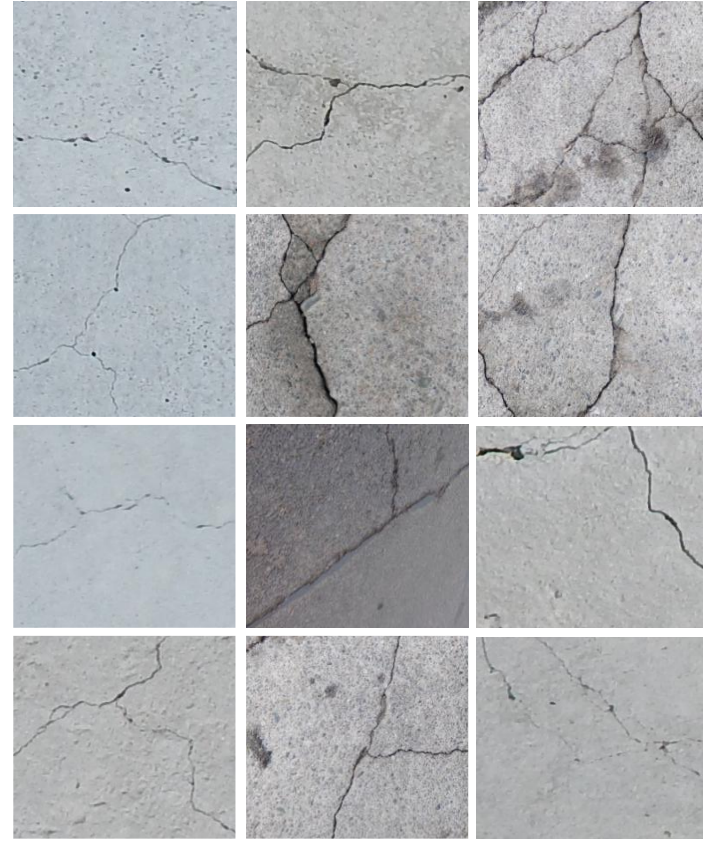


Fig. 4. Part of the training data diagram.

TABLE I

THE PARAMETERS OF ALL MODULES

Environmental parameters and hyperparameters	Value
Operating System	Windows 10
Deep Learning Framework	PyTorch
CPU	AMD Ryzen 5 5600U
Input Image Size	640×640
Training Epochs	200
Batch Size	16
Initial Learning Rate	0.001
Final OneCycleLR Learning Rate	0.2
Momentum	0.937
warmup epochs	3.0
Warmup Initial Momentum	0.8
Warmup Initial Bias	0.1
Optimizer Weight Decay	0.0005

B. Ablation experiment

To verify the effectiveness of adding GAM attention mechanism to the Backbone and introducing RepGFPN to the Neck as well as the advantages of improving model performance, ablation experiments were conducted to evaluate the training effects of the same data set under different models, and the results are shown in Table II.

TABLE II

RESULTS OF THE ABLATION EXPERIMENT

Network model	P (%)	R (%)	mAP (%)
yolov5s	33.7	52.5	33.2
yolov5s+gam	53.6	47.2	46.7
yolov5s+gam+regfpn (GR-YOLO)	57.6	51.4	51.0

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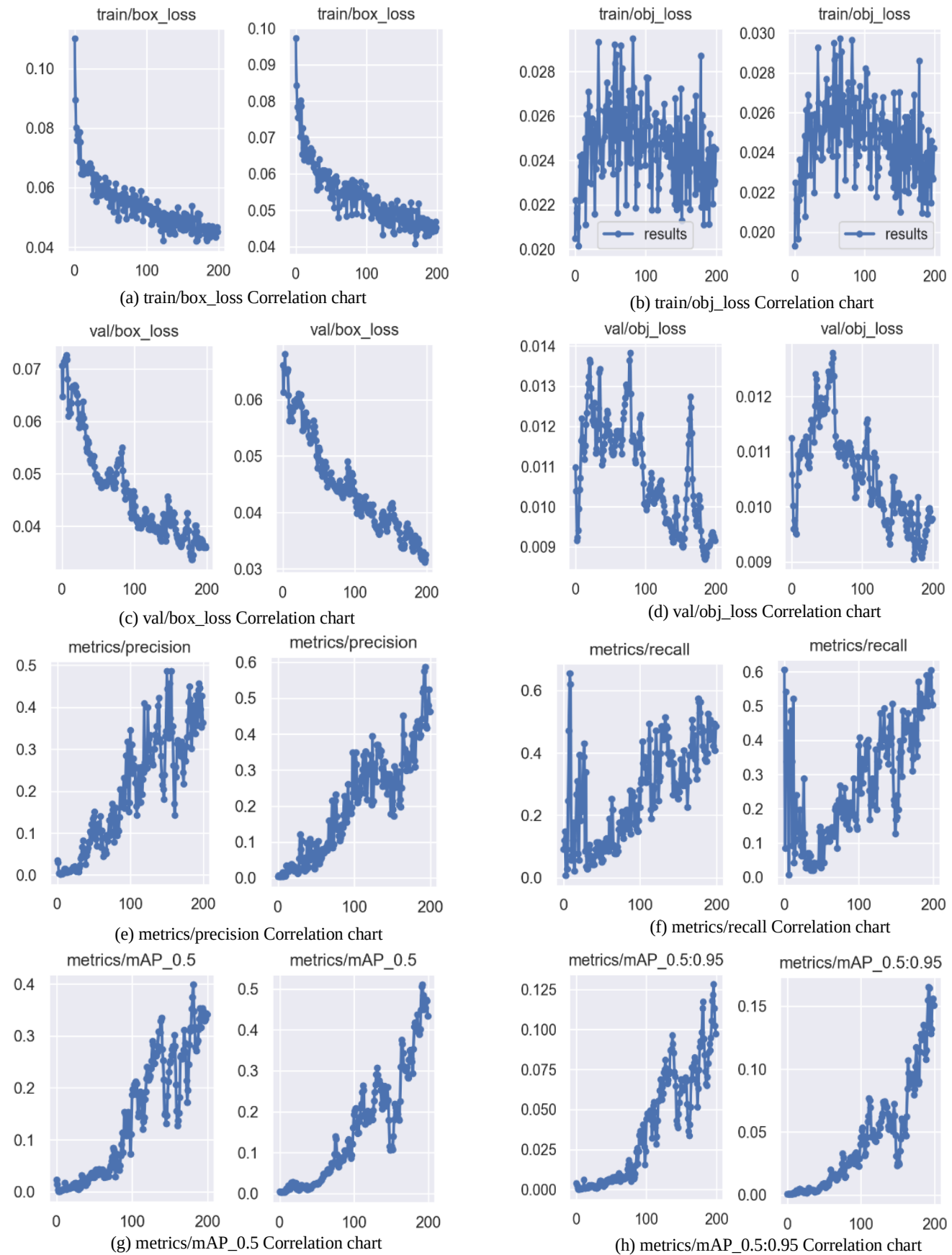


Fig. 5. Comparison of training results between yolov5s and GR-YOLO.

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Table II above demonstrates a significant enhancement in terms of accuracy P and average precision mAP with the proposed method. In comparison to the YOLOv5s model, there was a remarkable 23.9% increase in accuracy and an impressive 17.8% improvement in average accuracy. However, there existed a minimal disparity between the proposed method and YOLOv5s regarding recall rate, while the former exhibited substantial advantages concerning accuracy and average accuracy. Consequently, considering all aspects comprehensively, the proposed method outperformed other models presented in this table.

C. Contrast experiment

Fig. 5 shows the experimental results of yolov5s and GR-YOLO respectively, which include the training loss curve, accuracy P , recall rate R , and average accuracy mAP curve. The left side is the graph of yolov5s, and the right side is the graph of GR-YOLO. Since classification loss was 1 and classification loss was 0, classification loss was not compared.

As can be seen from Fig. 5(a) and Fig. 5(b), the training loss of GR-YOLO is the same as that of yolov5s. However, as can be seen in Fig. 5(c), the box regression loss (box_loss) of GR-YOLO is about 0.03 as the number of iterations increases, which is slightly better than that of yolov5. As can be seen in Figure 5(e), the accuracy of GR-YOLO is about 60%, which is about 10% better than that of yolov5s. Fig. 5(g) shows the accuracy of mAP when the Intersection over Union (IoU) is 0.5, and the mAP is improved by about 11% compared with the yolov5s. Fig. 5(h) shows the average mAP of IoU from 0.5 to 0.95 with a step size of 0.05, and the mAP of GR-YOLO is about 2.5% higher than that of yolov5s.

To better verify the accuracy and effectiveness of the proposed method, the experimental comparison under multiple working conditions was carried out. Compared with yolov5s, yolov8, S Tang et al., the results are as follows:

TABLE III

RESULTS OF THE CONTRAST EXPERIMENT

Network model	P (%)	R (%)	mAP (%)
yolov5s[31]	33.7	52.5	33.2
yolov8[32]	20.0	24.6	15.4
S Tang et al.[33]	17.0	18.3	10.0
yolov5s+gam+regfpn (GR-YOLO)	57.6	51.4	51.0

In Table III, the accuracy (P) of the proposed method is 57.6%, which is about 37.6% higher than that of yolov8 and about 40.6% higher than that of S Tang et al. The recall rate (R) of this method was about 26.8% and 33.1% higher than that of yolov8 and S Tang et al., respectively. The average accuracy (mAP) of this method was about 35.6% and 41.0% higher than that of yolov8 and S Tang et al., respectively.

To further verify the superiority of the proposed method compared with other methods, yolov5s, yolov8, S Tang et al. and GR-YOLO were selected to detect some wall cracks. The test results are shown in Fig. 6. It can be seen from Fig. 6 that the detection effect of the proposed method was better than the other three methods, which further indicated that the method had more advantages in detecting wall cracks.

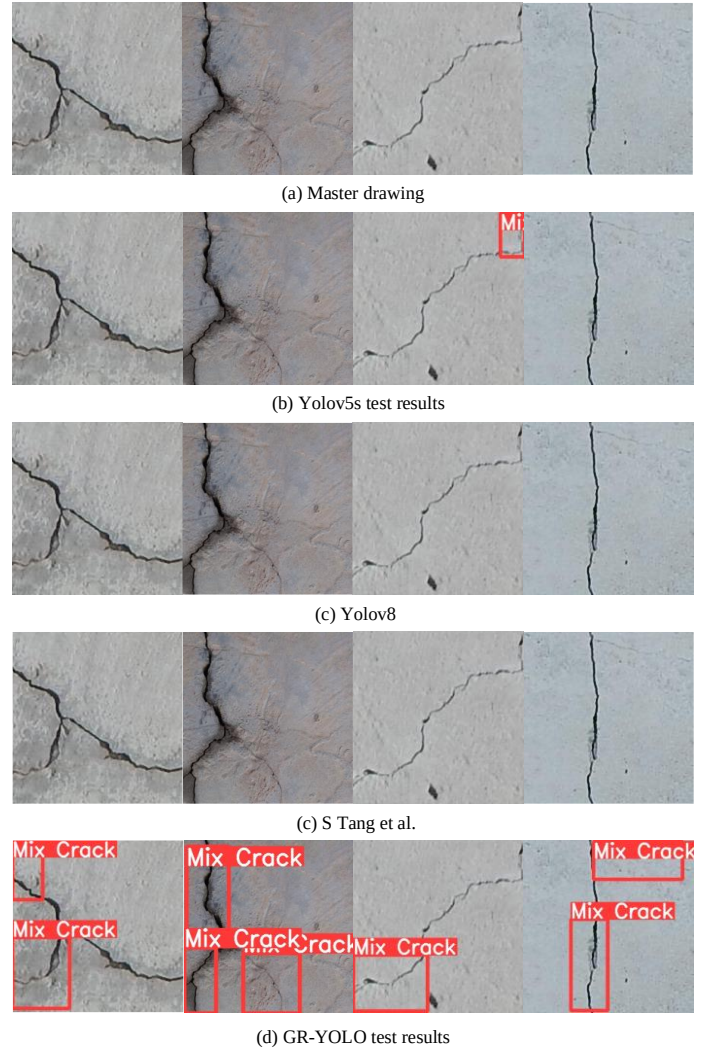


Fig. 6. Comparison of test results.

IV. DISCUSSION

This section provides a comprehensive analysis of the proposed method by comparing its advantages and limitations with those of existing approaches. It delves into how the method performs across a range of working conditions, diverse scenarios, and varying crack shapes, highlighting both its strengths and areas where it may fall short. The discussion also explores the specific challenges encountered in these different contexts, offering insights into the method's overall robustness and suggesting potential areas for improvement.

To explore the effects of the model and the number of iterations on the accuracy (P), recall rate (R), and average accuracy (mAP), the results of different methods under different iterations are shown in Fig. 7.

In Fig. 7 (b), the recall rate of yolov5s is higher than that of the method proposed in this paper, yolov5s oscillates continuously during the iteration process, and the result is very unstable. However, although the improvement of this method was very slow, it was still in a state of steady improvement. The recall rate R of yolov8 was higher in the early stage of training and decreased sharply in the later stage, which may be due to

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the over-fitting of the model. In Fig. 7 (a) and Fig. 7 (c), S Tang et al.'s results increase slowly with the increase of stacking times. YOLOv8 also had the same problem, while yolov5 and the method in this paper did not have this phenomenon. The method in this paper has steadily improved in accuracy P and average accuracy mAP, and the amplitude was greater than yolov5. It could be seen that the crack detection effect of yolov5 was better than that of yolov8 and S Tang et al., and the improved effect of this paper was due to yolov5.

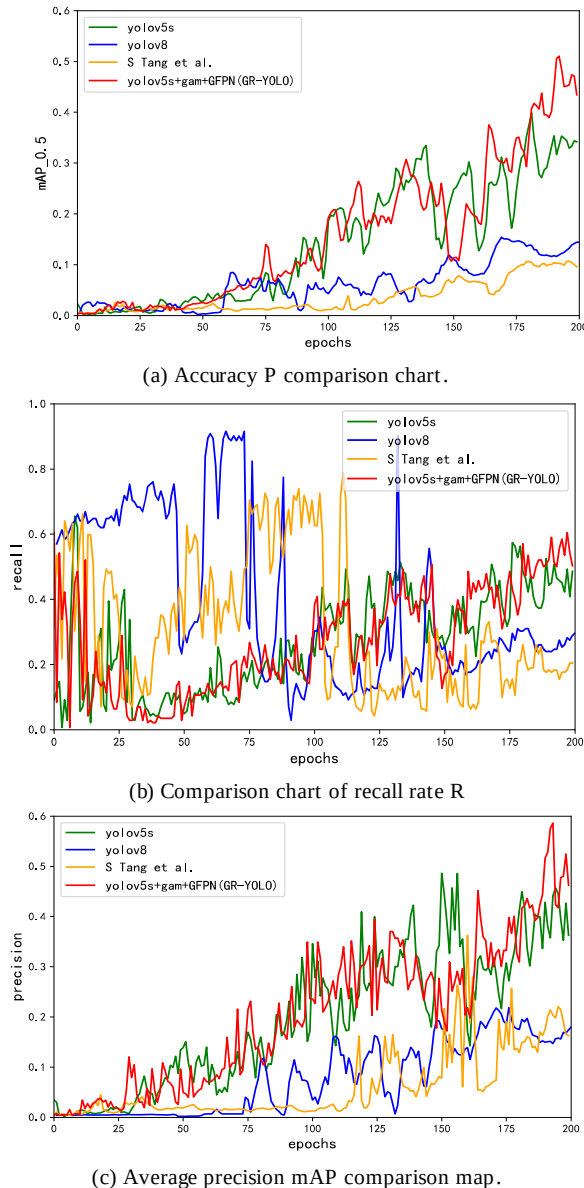


Fig. 7. Results of different methods at different iterations.

The performance and limitations of the proposed method under different working conditions, various scenarios, and different crack shapes. To further verify the universality of the proposed method, experiments were carried out under the conditions of wet walls, light occlusion, and different shooting angles, which also verified the effectiveness of the proposed method. The specific results are shown in Fig.8.

In the above diagram, it can be seen that the overall effect of the model is good, and most of the cracks can be detected, but

there are still some problems. Fig. 8 (a) is the detection effect of wet walls. Although most of the cracks are detected, there are some phenomena such as missed detection and repeated detection, and the recognition ability of some small cracks is poor. Fig. 8 (b) is the detection effect under light occlusion; in this environment, the model recognition effect is better. Fig. 8 (c) shows the detection effect at different shooting angles. The model had a general effect on the identification of some cracks with large inclination angles. All in all, the model was better for some cracks that were not obvious and difficult to detect by the naked eye, but it had a general effect on the identification of some small cracks and oblique cracks. This may have been because small cracks had a lower number of pixels, resulting in them being less obvious in the image, which required the algorithm to have higher resolution and more sensitive detection capabilities. Additionally, a large crack inclination angle might have led to difficulties in feature extraction and positioning, which were the directions we needed to improve in the future.

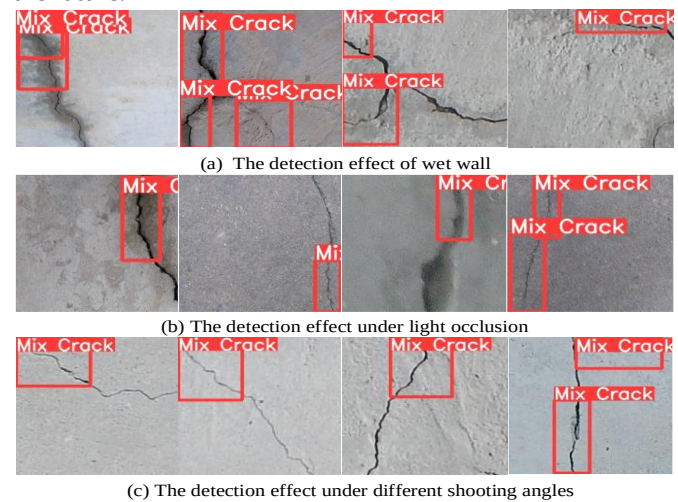


Fig. 8. Experimental verification diagram.

By analyzing the detailed parameters of each model structure, as shown in Table IV, it becomes evident that while the proposed method outperforms other approaches in terms of detection accuracy, it also comes with a significantly higher number of parameters. This increased complexity may pose challenges when deploying the model across various hardware platforms, particularly those with limited computational resources. Therefore, one of the key areas for future work will be to optimize and refine the model to make it more lightweight and efficient, without compromising its accuracy.

TABLE IV

DETAILED PARAMETERS OF THE NETWORK MODEL			
Network model	Layers	Parameters	GFLOPs
yolov5s	214	7022326	15.9
yolov8	168	3005843	8.1
S Tang et al.	285	9335494	31.2
yolov5s+gam+regfpn (GR-YOLO)	281	10980982	20.9

V. CONCLUSION

In this paper, a hybrid attention mechanism and RepGFPPN method for wall crack detection of a high-altitude cleaning

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robot have been proposed. To improve the accuracy and speed of the neural network, the GAM attention mechanism was incorporated to the YOLOv5 backbone network to effectively improve the accuracy of feature extraction. The introduction of RepFPN in the neck improved the ability to fuse multi-scale features, significantly improving crack detection and computational efficiency. The experimental results verified the effectiveness of the proposed method. Compared to the original YOLOv5 model, mAP improved by 13.5% after incorporating the GAM attention mechanism, and mAP improved by 17.8% after introducing RepGFPN. The improved model has notably enhanced crack detection accuracy and holds significant application potential.

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